

Inputs

selcal validate

selcal run

SQLite record

Series file

CSV or NPZ
source x, target y

Plan file (JSON)

statistic, lag set
selection rule
null model
 α , replicates B, seed

1 Input valid?

series loads, passes checks

exit 2

2 Plan executable?

resource budget within caps
smallest attainable $p \leq \alpha$

exit 2

else NOT_EXECUTABLE, e.g.
REFUSE_NULL_STATES_TOO_FEW

3 Scientific assumptions

declared, not verified;
the analyst must justify them

reads the same files and repeats preflight;
an unattainable plan is refused unless
--allow-unattainable-plan

exit 2

resolve_plan_v2

```
statistic  lagged_pearson_v1  
           equal_width_binned_nette_v1  
           circular_shift_exact_v1 (exact)  
           circular_shift_v2 (sampled)  
           block_shuffle_v2  
null
```

calibrate_selected_family

bind null; observed data: scan every lag,
select c^* , keep T_{obs}

each replicate $b = 1 \dots B$

transform series $\rightarrow$ scan all lags
 $\rightarrow$ reselect $\rightarrow T_b$; failures kept

$E = \#\{b : T_b \geq T_{\text{obs}}\}, \quad p = (1 + E) / (B + 1)$

CalibrationResult $\rightarrow$ verify_calibration_result

COMPLETE: p and decision
or NOT_EVALUABLE: no p

exit 7

one file, four members

input (raw bytes)

request (plan)

result

software identity

selcal verify

consistency check

selcal verify --replay

recompute; must MATCH

selcal report

HTML report

exit 7 if the stored result
is NOT_EVALUABLE

Where SelCal stops

exit 2

guard refuses before any calibration: invalid input, or plan
not executable (budget, unattainable p); nothing is recorded

exit 7

NOT_EVALUABLE: a failure (null binding, observed scan or any
replicate) is kept in the result; no p-value is reported